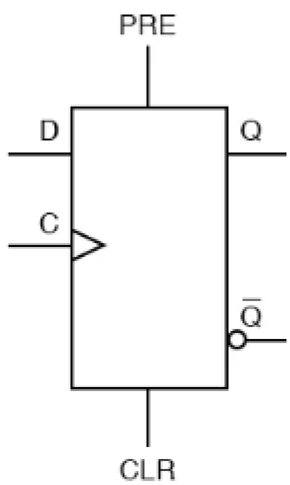
Circuit Simulation Lab:15
Design of Asynchronous Circuits

The purpose of this experiment is to introduce the design of asynchronous sequential circuits, in this case Asyn-D Flipflop. Asynchronous inputs on a flip-flop have control over the outputs (Q and not-Q) regardless of clock input status. These inputs are called the preset (PRE) and clear (CLR). The preset input drives the flip-flop to a set state while the clear input drives it to a reset state.



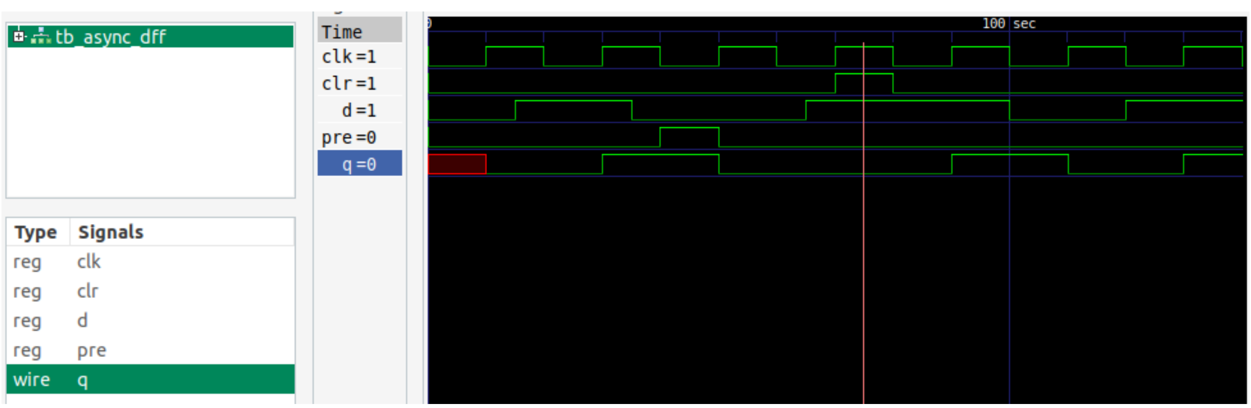

```
module async_dff(
input clk,
input d,
input pre,
input clr,
output reg q
);

always @(posedge clk or posedge pre or posedge clr)
begin

if(clr)
   q <= 0;

else if(pre)
   q <= 1;
```

```
else
    q <= d;

end

Endmodule
```

Testvbench

```
module tb_async_dff;

reg clk;
reg d;
reg pre;
reg clr;

wire q;

async_dff dut(
.clk(clk),
.d(d),
.pre(pre),
.clr(clr),
.q(q)
);

always #10 clk = ~clk;

initial begin

$dumpfile("async_dff.vcd");
$dumpvars(0,tb_async_dff);

clk = 0;
d   = 0;
pre = 0;
clr = 0;

// Normal DFF Operation
#15 d = 1;
#20 d = 0;

// Asynchronous Preset
```

```
#5  pre = 1;
#10 pre = 0;

// Normal Operation
#15 d = 1;

// Asynchronous Clear
#5  clr = 1;
#10 clr = 0;

// Normal Operation
#20 d = 0;
#20 d = 1;

#20;
$finish;

end

initial begin
$monitor("time=%0t clk=%b d=%b pre=%b clr=%b q=%b",
        $time,clk,d,pre,clr,q);
end

Endmodule
```



```
time=15 clk=1 d=1 pre=0 clr=0 q=0
time=20 clk=0 d=1 pre=0 clr=0 q=0
time=30 clk=1 d=1 pre=0 clr=0 q=1
time=35 clk=1 d=0 pre=0 clr=0 q=1
time=40 clk=0 d=0 pre=1 clr=0 q=1
time=50 clk=1 d=0 pre=0 clr=0 q=0
time=60 clk=0 d=0 pre=0 clr=0 q=0
time=65 clk=0 d=1 pre=0 clr=0 q=0
time=70 clk=1 d=1 pre=0 clr=1 q=0
time=80 clk=0 d=1 pre=0 clr=0 q=0
time=90 clk=1 d=1 pre=0 clr=0 q=1
time=100 clk=0 d=0 pre=0 clr=0 q=1
time=110 clk=1 d=0 pre=0 clr=0 q=0
time=120 clk=0 d=1 pre=0 clr=0 q=0
time=130 clk=1 d=1 pre=0 clr=0 q=1
time=140 clk=0 d=1 pre=0 clr=0 q=1
```